CLASS SYLLABUS · 2026–2027

Manufacturing Engineering Technology I

Design it, then make it real. Here's how our shop runs and what I expect.

Mr. Gavaldon · Da Vinci School
Contact: robertgavaldon@burnhamwood.org

DAILY

How this class runs

1. Grab your **assigned laptop**
2. **Sign in** to the program we're using
3. **Start work** — pick up where you left off
4. **Wait for instruction** when I call the class

PHONES

Up at the **start, every day.** Like the movies: phone away so everyone can focus. **I'll ask once** — then an email home, then a parent + admin meeting. Goal: **never** need it.

TUTORING & NOT FAILING

Fail a test → **one week of mandatory tutoring** (the department or me, on tutoring days). It's how we keep you from failing. Seats are assigned.

QUALITY OF WORK

- ✓ Finish your work **on your own**.
- ✓ Classwork done **in class**, the day it's assigned.
- ✓ On THIS class? No other classes or activities.
- ✓ **No AI in class or on tests** — you have to know it.

NOT ACCEPTED

- ✗ Yelling, fighting, hitting/touching others
- ✗ Eating, cussing, out of uniform
- ✗ Not turning in / not working in class
- ✗ No-showing tutoring; table-hopping; daily restroom

CORE 4 Respect Responsibility Quality of Work Quality of Self

Effort you can see.

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THE YEAR AT A GLANCE

Course Roadmap

Manufacturing Engineering Technology I
tentative – may shift

SEMESTER 1 · FALL

1 6-Wks
Aug – Sep

Foundations, Shop & Lab Safety
5w

Design notebook, safety non-negotiables, engineering design process

2 6-Wks
Oct – Nov

Manufacturing Systems & Processes
5w

Inputs → process → outputs; job/batch/continuous; planning & logistics

3 6-Wks
Nov – Dec

Fusion CAD for Manufacturing
5w

Model parts to make; Design → Make workflow

★ Semester 1 Final

SEMESTER 2 · SPRING

4 6-Wks
Jan – Feb

3D Printing & CNC Prototyping
6w

Prototype → production; tolerances, materials, quality checks

5 6-Wks
Mar – Apr

Assembly, Quality & Production
6w

Assemblies, drawings, and a prototype-to-production build

6 6-Wks
Apr – May

Capstone Build + Certification
6w

Capstone project + Autodesk ACU prep/exam

★ Final Exam

BRING EVERY DAY

- ✓ A pencil or pen
- ✓ A pencil sharpener
- ✓ A glue stick
- ✓ A folder just for this class
- ✓ A composition notebook
- ✓ Colored pencils or markers
- ✓ Scissors
- ✓ A mouse

For the class — bring one: hand sanitizer, a box of Kleenex, or disinfecting wipes.

iPads are allowed for class work. The first time I see texting, it goes away — and it does not come back out.

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PROGRAM OF STUDY

The 4-Year Engineering Pathway

CTE Levels 1–4 · Grades 9–12 · leading to certification

1 Grade 9

Principles of Applied Engineering INTRODUCTORY §127.781

Engineering design process, safety, and **Autodesk Fusion CAD from week 2** — project past project.

▲ Fusion part-modeling portfolio ◆ Starts ACU: Sketching · Part Modeling

2 Grade 10

Manufacturing Engineering Technology I INTERMEDIATE §127.813

Design → make: 3D printing/CNC, quality, and mechanical/electrical & automation systems.

▲ Prototype-to-production build ◆ Adds Assembly · Drawings · OSHA-10 (opt.)

3 Grade 11

Engineering Design & Problem Solving · InSPIRESS ADVANCED §127.785

Full design process on a real problem; **design a payload and compete** (InSPIRESS / UAH).

▲ InSPIRESS competition entry ◆ Sit the ACU — Fusion exam

4 Grade 12

Engineering Design & Presentation II CAPSTONE §127.784

Capstone: advanced system design + presentation; lead a competition-grade design cycle.

▲ InSPIRESS capstone + portfolio ◆ ACU earned · stretch ACP / dual credit

CERTIFICATION **Autodesk Certified User — Fusion**

Certiport · 35–40 Q · 50 min · pass 700/1000

Levels follow the Texas CTE Program of Study (Level 1 intro → Level 4 capstone). Course codes tentative — confirm with the campus CTE coordinator.

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